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Form Follows Function? Linking Morphological and Functional Polycentricity

Martijn Burger and Evert Meijers

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Abstract

Empirical research establishing the costs and benefits that can be associated with polycentric urban systems is often called for but rather thin on the ground. In part, this is due to the persistence of what appear to be two analytically distinct approaches in understanding and measuring polycentricity: a morphological approach centring on nodal features and a functional approach focused on the relations between centres. Informed by the oft-overlooked but rich heritage of urban systems research, this paper presents a general theoretical framework that links both approaches and discusses the way both can be measured and compared in a coherent manner. Using the Netherlands as a test case, it is demonstrated that most regions tend to be more morphologically polycentric than functionally polycentric. The difference is largely explained by the size, external connectivity and degree of self-sufficiency of a region's principal centre.

1. Introduction

Over the past 15 years, a vast academic and policy literature has emerged focusing on the concepts of 'polycentrism' and 'polycentric development'. Nevertheless, polycentric development remains one of the most versatile and 'fuzzy' concepts around (see Markusen, 2003), despite widespread calls for further conceptual clarification (Kloosterman and Musterd, 2001; Davoudi, 2003; Hague and Kirk, 2003; Turok and Bailey, 2004; Hoyler

et al., 2008; Lambregts, 2009). Polycentricity definitely ranks among those key terms that are employed loosely and in a variety of ways and, as Parr (2008) warns, this inevitably leads to imprecision and a loss of meaning. While the versatility of the concept may partly explain its persisting prominence—as it seems to hold something for everyone (Waterhout, 2002; Davoudi, 2007)—at the same time the Babel-like confusion surrounding the

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concept impedes academic progress. As regards polycentric development, progress would mean empirically establishing the actual merits of polycentric development as a strategy and establishing the environmental, economic and social consequences of a move towards polycentric urban systems (see for example, Kloosterman and Musterd, 2001; Parr, 2004; Turok and Bailey, 2004; Davoudi, 2007; Meijers, 2008a; Hoyler *et al.*, 2008; Vandermotten *et al.*, 2008; Lambregts, 2009; Meijers and Burger, 2010).

However, the calls for further clarification of the concept of polycentricity may give the wrong impression that conceptual and analytical clarification of the concept has not progressed over recent years. The contrary holds, and those calling for clarification can be partly credited for this. For instance, Lambregts (2009) makes a useful distinction between three related but yet distinct approaches to polycentricity. The first sees polycentric development as a normative planning strategy applied at metropolitan, national and transnational scales (see for instance, Albrechts, 2001; Davoudi, 2003; Waterhout *et al.*, 2005). The second considers polycentric development as a spatial process, resulting from the outward diffusion of (often higher-order) urban functions from major centres to smaller nearby centres (Kloosterman and Musterd, 2001; Hall and Pain, 2006). A third approach considers the spatial outcome of this process and in the literature we find a plethora of concepts describing the resulting spatial configuration of contemporary urban areas (see Meijers, 2005, for an overview). Although the labels of these concepts nearly all contain the word 'polycentric' in various connections to such territorial concepts as 'city', 'urban region', 'mega-city-region', 'metropolitan area' and 'global city region', in practice we find greatly diverging interpretations of what makes such territories polycentric, as well as diverging approaches to measuring polycentricity.

The most considerable difference of opinion in the debate rests on the question of whether polycentricity refers just to morphological aspects of the urban system or whether it should also incorporate relational aspects between the centres making up the urban system in question (Green, 2007; Meijers, 2008b). The morphological dimension, referred to as morphological polycentricity, basically addresses the size and territorial distribution of the urban centres across the territory and equates more balanced distributions with polycentricity (see for example, Kloosterman and Lambregts, 2001; Parr, 2004; Meijers and Burger, 2010). The relational dimension, referred to as functional polycentricity, takes the functional connections between the settlements into account and considers a balanced, multidirectional set of relations to be more polycentric (ESPON 1.1.1, 2004; Green, 2007; de Goei *et al.*, 2010). Proponents of the functional polycentricity approach generally claim that nodes without balanced relations would not form a polycentric system (ESPON 1.1.1, 2004). In fact, the strength and orientation of linkages between centres or cities could well be major components of an explanation of the performance of the urban system as a whole.

However, according to Hoyler *et al.* (2008, p. 1058), combining morphological characteristics and functional relations in one approach "contributes to a conflation of two analytically distinct dimensions of polycentricity". Naturally, a balance in the size distribution of centres does not necessarily imply that there are functional linkages between the different centres, let alone an equal distribution of these linkages and the existence of multidirectional flow patterns. Accordingly, in the contemporary literature on urban systems, morphological polycentricity and functional polycentricity are considered to be two different analytical concepts and relatively little effort has been made to connect these two trains of thought. In addition, it remains unclear why some systems are morphologically polycentric and

not functionally polycentric, or vice versa (see for example, Hall and Pain, 2006).

In this paper, we explore the relationship between morphological and functional polycentricity. We present a general theoretical framework rooted in urban systems research which indicates the interdependency between the degree of morphological polycentricity (balance in the size distribution or absolute importance of centres) and functional polycentricity (balance in the distribution of functional linkages or relative importance across centres). To do so, we need to take into account a number of related features of urban systems, which include the network density and openness of urban systems. In this, we build on other analytical approaches to functional polycentricity by disentangling the directionality of the functional linkages between centres from the degree of network formation between centres (i.e. network density). As well as examining the rather unknown relationship between morphological and functional polycentricity, this paper also links these concepts of polycentricity to the literature on central places and urban systems. This literature has faded somewhat into the background in the past two decades (Coffey *et al.*, 1998), but still has great relevance for understanding the concept of polycentricity. Using the Netherlands as a test subject, we show how the degree of morphological polycentricity and functional polycentricity within territorial units can be jointly evaluated. We will also explain why the degree of morphological polycentricity and functional polycentricity differs within territories.

The remainder of this paper is organised as follows. Given that morphological polycentricity can be linked to the balance in the distribution of the absolute importance of centres and functional polycentricity to the balance in the distribution of relative importance across centres, how the importance of centres is conceptualised and measured are crucial questions. This has been a core issue

in classical central place studies and urban systems theory and section 2 discusses this literature. This discussion results in a theoretical model for studying morphological and functional polycentricity that will be applied in the case study presented in this paper. However, first, section 3 synthesises the literature on both approaches to polycentricity. Section 4 presents the research approach adopted in our empirical analysis, which itself is presented in section 5. This section compares morphological and functional polycentricity and explains the differences found using our theoretical model. Section 6 concludes with a discussion of the findings.

2. Conceptualising the Importance of Centres

2.1 Central Place Theory and the Importance of Centres

The study of the organisation of urban systems in urban geography, regional science, urban economics and spatial planning originates from urban location theory and can be traced back to the work of Christaller (1933) and Lösch (1944) on central place systems. Central place theory is occupied with the study of the distribution, size and number of cities and towns (Berry and Parr, 1988) and originally focused on urban–rural relationships, where the scope of interactions was most often confined to consumer-oriented trade (Berry and Pred, 1965). In a central place system, there is a hierarchy of central places, where the centrality of a settlement and the variety of goods and services it provides are thought to be perfectly correlated. Accordingly, lower-order central places are dependent on higher-order central places for the provision of goods and services and only a small proportion of the central places will be self-sufficient in that they offer the full range of goods. In this, lower-order centres do not provide goods and services to the highest-order central place and trade

between centres of similar size is considered redundant as these centres provide the same goods and services. Although the central place model does not officially say anything about journey-to-work flows as it was originally concerned with trade between centres, it can be expected that in a central place system the centre is characterised by an excess labour demand and the small places by an excess labour supply (Parr, 1987).

The central place model focuses on rural areas in general and city–hinterland relationships in particular and is, above all, a very specific theory about the spatial organisation of the local economy. However, the idea of a hierarchical urban system can be made more general in both theory and application and translated to higher spatial scales. In this, the literature that has built on central place theory has followed two different paths (McPherson, 1981). On the one hand, a number of economic studies have extended and modified the formal model, to arrive at a more general and realistic model of a hierarchical urban system (see Berry and Parr, 1988).¹ On the other hand, empirical research, mostly originating from the urban systems school (see Bourne and Simmons, 1978), has viewed central place studies in a more analytical way, without the restrictions of formal theory. In this, the goal was to explore the organisation of urban systems and try to understand the nature of city–hinterland relationships (Berry, 1964; Berry and Pred, 1965; Haggett, 1965). Drawing on general systems theory (von Bertalanffy, 1950), any urban system can be thought of as consisting of a set of interdependent nodes (for example, centres) and the patterns of interaction between these nodes (for example, trade, commuting) (Berry, 1964; Simmons, 1978). Central place theory predicts that all urban systems are by definition rather monocentric, given the emphasis put on the hierarchy, and not balance, of the importance of centres. Such a monocentric urban system can be perceived

as a nodal area containing a principal centre and several surrounding subordinate centres of different hierarchical orders that are part of the principal centre's market area (Haggett, 1965). Such an urban system is characterised by a hierarchy of centres that is rank-ordered on the basis of the size of their market areas and their complexity in terms of the number of functions provided (Berry and Garrison, 1958; Davies, 1967). From a network point of view, such a monocentric urban system is best represented by a star-shaped pattern of interactions, where the flows of goods, services and commuters between centres of different hierarchical orders are one-sided and centralised (Nystuen and Dacey, 1961; Haggett and Chorley, 1967).²

However, the hierarchical central place model, with its emphasis on monocentricity, has increasing difficulty explaining spatial reality (Batten, 1995; Coffey *et al.*, 1998; Meijers, 2007). One of the reasons is its inability to deal with the more polycentric spatial organisation of metropolitan areas that appears to be inherent to the post-industrial era and that is fuelled by globalisation (Kloosterman and Musterd, 2001; Scott *et al.*, 2001; Phelps and Ozawa, 2003). In other words, hierarchy appears to be a less dominant feature of many urban systems at all spatial scales.

2.2 Nodality versus Centrality in Urban Systems

Following Preston (1971, 1975), it is possible to distinguish between the absolute importance of a centre or its nodality and the relative importance of a centre or its centrality. Whereas the nodality of a centre can be expressed by its size and the range of functions it offers (Lukermann, 1966), the centrality of a centre is typically defined as the part of its importance that can be ascribed to the provision of goods, services and jobs in excess of those demanded by the centre's own inhabitants (Ullman, 1941; Preston 1971;

Barton, 1978; Marshall, 1989). This distinction goes back to the work by Christaller (1933). In his seminal work *Die Zentralen Orte in Süddeutschland*, it is argued that if the importance of a centre is only based on its size, then part of its importance must be ascribed to the settlement itself as an agglomeration and another part to the settlement as a central place, providing goods, services and jobs to surrounding places. Hence, it is desirable to separate the external importance from the local importance of a centre. The centrality of a centre c in a closed system of cities can then be defined as follows

$$C_c = N_c - L_c$$

in which, C_c = the surplus of importance of a centre based on incoming flows from other places—i.e. the relative importance of a centre, its *centrality*; N_c = the absolute importance of a centre based on internal and incoming external flows—i.e. its *nodality*; and L_c = the local importance of a centre based on *internal flows*.

To illustrate, when examining the importance of a centre as a job provider, it can be argued that N_c represents total employment in centre c , C_c represents the number of incoming commuters in centre c , and L_c represents the number of employees in centre c that also live there (see also Burger *et al.*, 2011). In a similar fashion, it is possible to look at shopping and producer-oriented trade.

Christaller's and Preston's distinction between nodality and centrality is entrenched in a much broader discussion in urban research that dates back to the 1950s and 1960s, which dealt with the question of whether the configuration of urban systems in general and the importance of central places in particular should be evaluated on the basis of the internal characteristics of centres or the external relations of centres.³ Although Christaller (1933) originally rank-ordered central places based on the external relations of centres, this practice was replaced by more broad and less restrictive

characterisations of functional aggregate importance—internal characteristics of centres—in post-war extensions and modifications of classical central place theory. In this, the explicit distinction between the local and extra-local importance of a settlement gradually got lost (Preston, 1971, 1975).

The motivation for this shift in focus came from both a theoretical and an empirical point of view. On the one hand, formal theoretical accounts of hierarchical spatial structure now related central place and market hierarchies to the distributions of city size (Beckmann, 1958; Parr, 1969; Beckmann and McPherson, 1970). On the other hand, there was a lack of data regarding the functional interaction between centres based on consumer, firm and commuting behaviour (Thompson, 1974). Hence, the number of studies that have measured the importance of cities on the basis of spatial interaction between centres has been, up until the end of the 1990s, relatively limited (Coffey *et al.*, 1998). Nevertheless, the question as to whether the most populous centres are also the most central centres in a system of cities continued to be challenged (see for example, Preston, 1975; and, more recently, Short, 2004; and Limtanakool *et al.*, 2007).

It is not difficult to draw parallels between this debate and the contemporary debate on morphological versus functional polycentricity. The discussion in this contemporary debate is also about measuring the importance of centres on either internal characteristics or on the basis of flows. Furthermore, good data on flows are still difficult to obtain. For instance, the ESPON 1.1.1 project (ESPON 1.1.1, 2004) approximates functional polycentricity by using an internal characteristic of cities—namely, their accessibility.

2.3. The Importance of Centres, Openness and the Spatial Scope of Activities

So far, we have considered the importance of centres in a closed or isolated urban system.

Accordingly, the centrality of a centre is determined on the basis of the surplus of importance within an urban system (for example, city-region or metropolitan region), where the surplus of importance derived from linkages with centres outside this system is ignored. This is at least to some extent problematic as contemporary urban systems are not entities that operate on their own and, certainly in the present-day economy, most urban systems interact at least to some extent. In this, it can be expected that centres at the top of the urban hierarchy in an urban system are disproportionally connected to this 'outside world' because of better accessibility and the higher-order functions they provide. Indeed, some centres fulfil a global or national function, while other centres fulfil a more regional or local function (Lambregts, 2009; Wall, 2009).

Extending Christaller's definition of centrality,⁴ the surplus of importance of a centre C_c within an urban system—for example a city-region or metropolitan region—can be thought to consist of a within-system component C_{ci} and an outside-system component C_{ce}

$$C_{ci} = N_c - C_{ce} - L_c$$

where, C_{ci} = the surplus of importance of a centre based on incoming flows from other places within the same urban system—its *internal centrality*; N_c = the absolute importance of a centre—its *nodality*; C_{ce} = the surplus of importance of a centre based on incoming flows from other places outside the urban system—its *external centrality*; and L_c = the local importance of a centre based on internal flows.

To illustrate, when examining the importance of a centre as a provider of employment in a city-region, it can be argued that N_c represents the total employment in centre c ; C_{ci} represents the incoming commuting in centre c from places situated within the city-region; C_{ce} represents the incoming commuting in centre c from places situated outside the

city-region, and L_c represents the number of employees in centre c that also live there. In this, C_{ci} and C_{ce} add up to the total centrality of a centre (see also Preston, 1971, 1975).

In the remainder of this paper, we make use of this extended model when analysing the relationship between morphological and functional polycentricity. In this, we will look at spatial structure at the intraurban or supralocal scale (see Kloosterman and Musterd, 2001) based on journey-to-work and consumer travel flows.

3. The Morphological and Functional Approaches to Polycentricity

In analogy with the distinction between nodality and centrality discussed in the previous section, two main approaches to measuring the spatial structure of systems can be distinguished: the morphological approach and the functional approach (Green, 2007; Meijers, 2008b).

3.1 Morphological Polycentricity

As exponents of the morphological approach to polycentricity assert, the term polycentricity basically refers to the plurality of urban centres in a given territory. However, polycentricity tends to be more closely associated with a *balanced distribution* with respect to the importance of these urban centres (see for example, Kloosterman and Lambregts, 2001; Parr, 2004; Meijers, 2008b). This interpretation is most probably inspired by existing policy debates at the national and the European scales, in which polycentricity is linked to the rise in importance of metropolitan areas relative to one or two existing core metropolitan areas. Thus, the main promise of the concept of polycentric development appears to be its ability to link the seemingly conflicting objectives of cohesion and competitiveness (Waterhout, 2002), a combination that is, however, far from evident (Krätke, 2001;

Meijers and Sandberg, 2008; Vandermotten *et al.*, 2008). Having studied the interpretation of polycentric development in policy strategies in European countries, Meijers *et al.* define a polycentric development policy as

a policy that addresses the distribution of economic and/or economically relevant functions over the [spatial] system in such a way that the urban hierarchy is flattened in a territorially balanced way (Meijers *et al.*, 2007, p. 7).

This lack of hierarchy in terms of size or absolute importance among the larger centres is also stressed by Parr (2004) and Kloosterman and Lambregts (2001) as a defining characteristic of a polycentric system at the regional scale. In other words, we have to distinguish 'polycentric' from concepts such as 'multi-centric' or 'multinuclear', the difference being that polycentricity puts more emphasis on the balanced distribution in the size of the multiple centres in an urban system.

3.2 Functional Polycentricity

Those who adhere to the relational or functional dimension of polycentricity do not dismiss the morphological approach, but rather, extend it to include also the pattern of functional interaction between the urban centres. The approach generally taken has many similarities with the morphological approach. Again, it is not so much about the existence or strength of functional relationships between centres, but rather about the balance in the distribution of the functional relationships. The more evenly flows are distributed between the centres—or, in other words, the more multidirectional rather than mono-directional (ESPON 1.1.1, 2004) they are—the more polycentric. Such an equal balance in the distribution of inflows can be found in an urban system in which functional relationships are not directed at one centre, but two-sided (reciprocal) and criss-cross

(also existing between smaller centres) (van der Laan, 1998; de Goei *et al.*, 2010; Burger *et al.*, 2011).⁵ In a recent, seminal contribution on functional polycentricity, Green (2007; see also, Green, 2008) adds another dimension, which is network density. The degree of network density reflects the extent to which centres in a region are functionally interdependent (that is open or not self-sufficient) and can be conceptualised as the ratio of the actual connections between centres to the total of potential connections between centres (Green, 2007). In our case, the total potential connections between the centres within a region can be defined as the sum of the absolute importance of the centres within a region or, alternatively, the sum of people working/shopping within a region. A generally low ratio of the sum of internal centrality scores of centres within a region to the sum of the absolute importance of centres within a region indicates a low level of network density.

3.3 Polycentricity, Nodality and Centrality

Important to note here is the link between nodality, centrality and both forms of polycentricity. As nodality and centrality reflect the absolute and relative importance of centres in an urban system, then morphological polycentricity and functional polycentricity should be about the balance in the absolute and relative importance of these centres. Hence, it can be argued that in a morphologically polycentric system there is no dominant centre or, alternatively, that centres are relatively equal in terms of nodality or their absolute importance. In a functional polycentric system, there is no dominant city—in other words, the relationships have no obvious orientation towards a particular centre; centres are relatively equal in terms of centrality or their relative importance (see Figure 1). Consequently, nodality provides the basis for measuring morphological polycentricity, whereas the measurement of functional

polycentricity is to be based on centrality. In existing analyses, most often attention is paid to the distribution of intraregional flows and hence internal centrality scores (Hall and Pain, 2006; Green, 2007; Burger *et al.*, 2011).

Note that we explicitly disentangle the degree of functional polycentricity (balance in the distribution of functional linkages) from the degree of network density (extent to which the centres are functionally linked). Not including network density in our measure of functional polycentricity (based on centrality) is necessary as it is possible to come across urban systems with a high network density, but which are hierarchically organised, and urban systems with a low network density, but in which centres are relatively equal in terms of their connectivity to other centres (see Figure 2). If both centralisation and network density scores are combined, we may find that urban systems with a highly unbalanced distribution of functional linkages but a high network density would receive a similar score to those urban systems with a highly balanced distribution of functional linkages but a low network density. In fact, the perhaps remarkable finding in the POLYNET study (Hall and Pain, 2006)—that a morphologically

monocentric region such as Greater London is more functionally polycentric than morphologically polycentric regions such as central Belgium and northern Switzerland—can probably be ascribed mainly to the lack of network formation between the centres in the latter regions. Hence, we argue that, for conceptual clarification and in conformity with common practice in network analysis (Wasserman and Faust, 1994),⁶ it is better not to equate a functionally polycentric urban system with a networked urban system. This does not mean that the degree of network density is not an important aspect of the organisation of a spatial system. In actual fact, synergies between the centres in an urban system will not be achieved without linkages between them (Meijers, 2005) and, within a policy context, one cannot speak of a functionally integrated urban region without linkages resulting from economic complementarities between the different centres (van Oort *et al.*, 2010). Finally, separating functional polycentricity and network density also facilitates comparison with morphological polycentricity. Obviously, the distributions of local importance and external centrality provide starting-points to explain the difference between morphological and functional polycentricity. In the next sections, we present our empirical assessment of the relationship between both forms of polycentricity. Section 4 presents the case study regions, the research approach and the data. Section 5 presents the analysis.

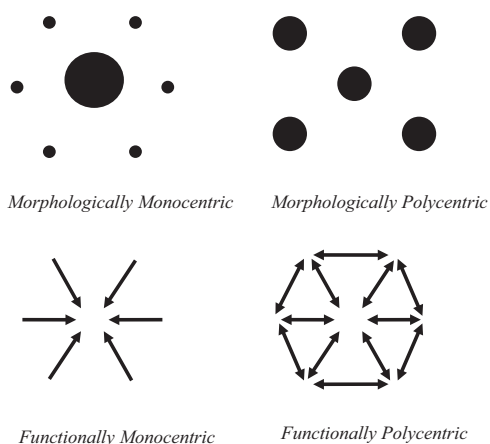


Figure 1. Morphological polycentricity versus functional polycentricity.

4. Case Study: Polycentricity in Dutch WGR Regions

4.1 WGR Regions in the Netherlands

The framework developed in the previous section will be applied to the Netherlands. While we could have taken any country, the Netherlands is of particular interest as it is widely known that polycentricity is a key characteristic of its spatial organisation

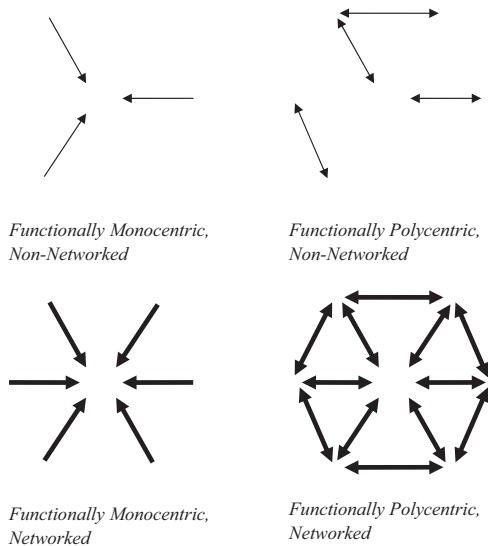


Figure 2. Functional mono/polycentric systems versus networked systems.

(Lambregts, 2009). The conceptual framework we presented is quintessentially scale-free and hence can be applied to any spatial entity ranging from individual cities to continents. Here, we decided to apply the model to 42 functionally coherent regions that together cover the entire Netherlands. These regions are referred to as ‘WGR’ regions and they get their name from the Intermunicipal Statutory Regulations Act (‘Wet Gemeenschappelijke Regelingen’—WGR) that enables municipalities to work jointly on issues that need to be addressed at a higher spatial scale than the municipal scale by means of issue-based common agreements. The Act does not specify which issues should be jointly addressed, but in practice these often concern regional aspects of economic development, tourism, recreation, housing, employment, traffic and transport, spatial development, nature and environmental affairs, welfare and social affairs. As the delimitation of WGR regions is based on municipal and provincial administrators’ and councillors’ perceptions of the scale at which issues in need of a regionally

co-ordinated approach arise, these regions provide an indirect proxy of functionally coherent regions. Despite the ‘professional’ definition of this region, the outcome appears generally well defensible, coinciding fairly well with what are believed to be travel-to-work areas and, consequently, has not led to a great debate on its rationality.⁷ Figure 3 presents these 42 regions. In the remainder of this paper, we refer to these regions by the name of their largest centre. Note that we collected data on the nodality and centrality of the four largest cities or towns in these regions.

4.2 Quantifying Spatial Structure

As explained in the previous sections, polycentricity is all about the balance in importance of urban centres. The more even the importance in terms of nodality and centrality of urban centres, and hence the less hierarchy, the more morphologically and functionally polycentric the system is. The rank–size distribution with regards to the importance of cities provides information on this hierarchy of centres and is therefore a good measure of the degree of monocentricity or polycentricity (Parr, 2004; Spiekermann and Wegener in ESPON 1.1.1, 2004; Meijers, 2008b; Adolphson, 2009). We adhere to this view and use the rank–size distribution of the nodality scores in an urban system to assess the degree of morphological polycentricity and the rank–size distribution of the centrality scores in an urban system to assess the degree of functional polycentricity. The major indicator is the slope of the regression line that best fits these rank–size distributions. The flatter the slope of this line is, the more polycentric the region. Conversely, the steeper the slope of this line is, the more monocentric the region.

As Meijers (2008b) points out, a crucial question concerns the number of urban centres ranked in the rank–size distributions. The extent of monocentricity or polycentricity is generally judged on the basis of the

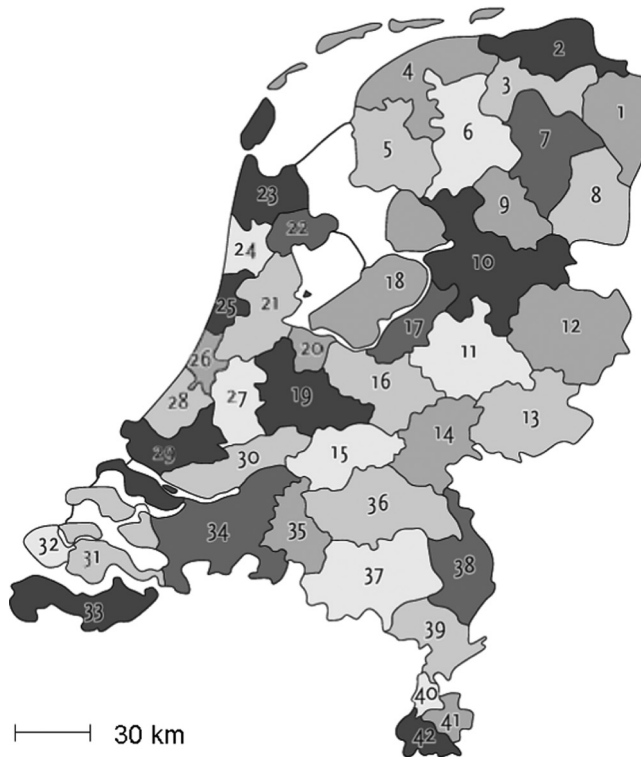


Figure 3. WGR regions in the Netherlands.

Source: CBS.

Key: 1 Oost-Groningen (Veendam); 2 Noord-Groningen and Eemsmond (Delfzijl); 3 Centraal and West. Groningen (Groningen); 4 Friesland Noord (Leeuwarden); 5 Zuidwest-Friesland (Sneek); 6 Friesland-Oost (Drachten); 7 Noord- and Midden-Drenthe (Assen); 8 Zuidoost-Drenthe (Emmen); 9 Zuidwest-Drenthe (Hoogeveen); 10 IJssel-Vecht (Zwolle); 11 Stedendriehoek (Apeldoorn); 12 Twente (Enschede); 13 Oost-Gelderland (Doetinchem); 14 Arnhem-Nijmegen (Nijmegen); 15 Rivierenland (Tiel); 16 Eem and Vallei (Amersfoort); 17 Noordwest-Veluwe (Harderwijk); 18 Flevoland (Almere); 19 Utrecht (Utrecht); 20 Gooi and Vechtstreek (Hilversum); 21 Aggl. Amsterdam (Amsterdam); 22 Westfriesland (Hoorn); 23 Kop Noord-Holland (Den Helder); 24 Noord-Kennemerland (Alkmaar); 25 West-Kennemerland (Haarlem); 26 Zuid-Holland-Noord (Leiden); 27 Zuid-Holland-Oost (Gouda); 28 Haaglanden ('s-Gravenhage); 29 Rijnmond (Rotterdam); 30 Zuid-Holland-Zuid (Dordrecht); 31 Oosterschelderegio (Goes); 32 Walcheren (Middelburg); 33 Zeeuwsch-Vlaanderen (Terneuzen); 34 West-Brabant (Breda); 35 Midden-Brabant (Tilburg); 36 Noordoost-Brabant ('s-Hertogenbosch); 37 Zuidoost-Brabant (Eindhoven); 38 Noord-Limburg (Venlo); 39 Midden-Limburg (Roermond); 40 Westelijke Mijnstreek (Sittard); 41 Oostelijk Zuid-Limburg (Heerlen); 42 Maastricht and Mergelland (Maastricht)

nodality and internal centrality of just the handful of largest cities. In general, sample size can be based on a fixed number of cities, a fixed size threshold or a size above which the sample accounts for some given proportion of a region's total nodality or internal

centrality (Cheshire, 1999). The latter method has disadvantages, as it is apparent that the number of centres included in the analysis is large for polycentric systems and small for monocentric systems. Hence, the number of centres including some given proportion of

the nodality or centrality is in itself an indicator of monocentricity or polycentricity and applying such a measure twice would distort the picture. A fixed size threshold is equally less appropriate as, in large and more densely populated urban systems, a centre of say 5000 inhabitants may be insignificant, while it could be of considerable absolute and relative importance in small or less populated systems. Hence, when measuring morphological and functional polycentricity on the basis of the rank–size distribution, the sample size could best be based on a fixed number of centres. In line with Meijers and Burger (2010), we used different numbers of places per region (2, 3 and 4) and then calculated the average of these three scores.

Figure 4 presents the four largest places (in terms of employment) in each of two Dutch regions (Maastricht and Sittard-Geleen) including the regression lines that best fit the rank–size distribution.⁸ In this example, Maastricht is obviously a morphologically monocentric region, while Sittard–Geleen is a clear example of a morphologically polycentric region. This brings us to an important issue that needs to be taken into account when analysing the results and figures provided later. This is that, in our texts and figures, we refer to the degree of polycentricity. However, as can also be seen in Figure 4, our measure based on the rank–size distribution

positions regions on a scale ranging from very monocentric to very polycentric. So, regions with a low level of polycentricity are actually monocentric and only regions with a high level of morphological polycentricity can be truly considered polycentric urban regions as addressed by authors such as Champion (2001), Kloosterman and Musterd (2001), Parr (2004), van Oort *et al.* (2010) and Cowell (2010).

4.3 Data

To examine the relationship between morphological and functional polycentricity, we estimated the slope of the regression line of the rank–size distribution of the nodality and internal centrality scores of the largest places in all 42 WGR regions (see Figure 3). More specifically, the nodality scores are used to assess the degree of morphological polycentricity and the internal centrality scores are used to assess the degree of functional polycentricity. We performed two analyses, one on the basis of commuting and one on the basis of shopping trips. We based both the nodality and the internal centrality scores on these trips. These flow data are drawn from the Dutch National Travel Survey 2004–08 (Mobiliteitsonderzoek Nederland).⁹ As indicated in the previous sections, the degree of nodality of a place is determined on the basis of employment (i.e. total incoming

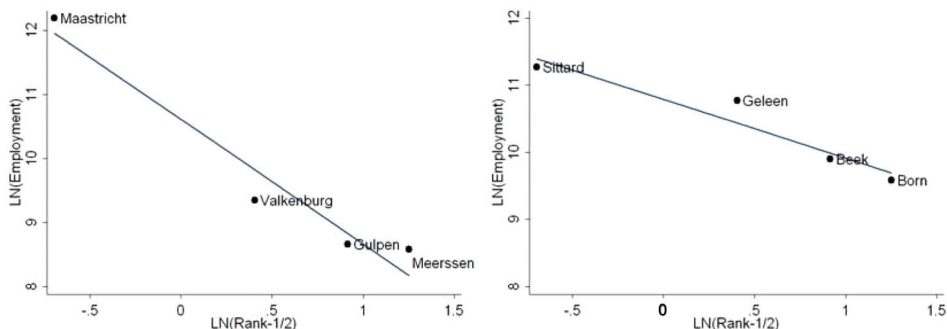


Figure 4. Rank–size distributions to measure mono/polycentricity.

journey-to-work flows, including those flows originating from its own centre as well as the places situated outside the WGR region) and the total number of shoppers. Likewise, the internal centrality of a place is determined on the basis of the total incoming journey-to-work and shopping flows from places situated within the same WGR region.

5. Empirical Analysis of Polycentricity in Dutch WGR Regions

5.1 Comparing Morphological and Functional Polycentricity

Table 1 shows the difference between the degree of morphological and functional polycentricity in Dutch WGR regions based on commuting and shopping respectively. A number of conclusions can be drawn. First, spatial structure differs across regions. Some regions are predominantly monocentric while other WGR regions are predominantly polycentric and most city-regions are somewhere in between. Overall, similar patterns can be observed for commuting and shopping.

Secondly, although there is a considerable correlation between the degree of morphological and functional polycentricity (0.84 for commuting and 0.57 for shopping; see Table 2), almost all regions are relatively more functionally polycentric than morphologically polycentric. For both commuting and shopping, the distribution of incoming flows from places located within the WGR region is more equal than the size distribution of centres.

These results differ somewhat from the POLYNET study (Hall and Pain, 2006), in which it was found that morphologically polycentric regions are not necessarily functionally polycentric and the degree of morphological polycentricity is generally stronger than the degree of functional polycentricity. However, in the POLYNET study, functional polycentricity was measured by an index containing both

the balance in the distribution of linkages and network density.

Table 2 (employment and shopping) indicates the relationship between the different aspects of the organisation of spatial systems. From this table, it can be obtained that the more morphologically polycentric a region is, the higher the network density, although this relationship is not very strong. Network density here is measured as the ratio between internal centrality and nodality—in other words, the ratio between flows in the region and total employment. The higher this ratio, the more strongly networked the cities in the region. Hence, the more morphologically polycentric a region is, the higher the degree of network formation between the cities. However, network density is not related to the balance in the directions of commuting flows (functional polycentricity), which provides support for our viewpoint that both should be disentangled. Finally, a POLYNET finding is that intraregional connectivity is often less hierarchical than external connectivity (Hall and Pain, 2006; Lambregts, 2009). This is confirmed by our findings, as the intraregional distribution of commuting flows (functional polycentricity) tends to be more balanced than the distribution of external centrality (see Figure 5).¹⁰

5.2. The Difference between Morphological and Functional Polycentricity

Recall that the nodality of a centre is the sum of its internal centrality, external centrality and local importance (section 2.3)—or, in other words, the total of flows (commuting or shopping) directed at this centre. Consequently, the difference between the degree of morphological and functional polycentricity can be explained by two factors: the distribution of local importance (extent to which flows remain within the same city) and the distribution of external centrality (extent to which the cities receive flows from outside

Table 1. Morphological polycentricity (MP) versus functional polycentricity (FP) in Dutch WGR regions

Region	Employment			Shopping		
	MP	FP	MP-FP	MP	FP	MP-FP
Veendam	-0.31	-0.47	0.16	-0.22	-0.53	0.31
Delfzijl	-0.28	-0.24	-0.03	-0.36	-0.29	-0.07
Groningen	-1.95	-1.13	-0.82	-1.70	-0.76	-0.93
Leeuwarden	-1.60	-1.34	-0.26	-1.22	-0.67	-0.55
Sneek	-0.91	-1.02	0.10	-0.76	-0.70	-0.06
Drachten	-0.73	-0.63	-0.10	-0.56	-0.50	-0.05
Assen	-1.13	-1.04	-0.09	-1.10	-0.71	-0.39
Emmen	-1.42	-1.23	-0.18	-1.22	-1.06	-0.15
Hoogeveen	-1.09	-0.71	-0.38	-0.89	-0.88	-0.01
Zwolle	-1.35	-1.24	-0.10	-1.08	-0.44	-0.64
Apeldoorn	-0.84	-0.58	-0.27	-0.80	-0.48	-0.32
Enschede	-0.45	-0.21	-0.23	-0.48	-0.11	-0.37
Doetinchem	-0.87	-0.85	-0.01	-0.73	-0.56	-0.16
Nijmegen	-0.61	-0.79	0.17	-0.54	-0.36	-0.18
Tiel	-0.42	-0.28	-0.14	-0.57	-0.44	-0.13
Amersfoort	-0.76	-0.75	-0.02	-0.71	-0.64	-0.07
Harderwijk	-0.39	-0.28	-0.11	-0.39	-0.21	-0.18
Almere	-0.87	-0.74	-0.13	-0.94	-0.46	-0.49
Utrecht	-1.30	-1.17	-0.13	-1.13	-0.90	-0.23
Hilversum	-0.96	-0.66	-0.30	-0.73	-0.58	-0.14
Amsterdam	-1.51	-1.04	-0.47	-1.46	-0.72	-0.74
Hoorn	-0.94	-0.72	-0.22	-1.05	-0.62	-0.43
Den Helder	-1.21	-0.46	-0.75	-0.94	-0.91	-0.03
Alkmaar	-1.15	-1.12	-0.02	-0.83	-1.18	0.35
Haarlem	-1.00	-0.51	-0.49	-0.87	-0.32	-0.55
Leiden	-1.08	-0.69	-0.40	-0.79	-0.59	-0.20
Gouda	-0.52	-0.55	0.04	-0.49	-0.25	-0.24
's-Gravenhage	-1.27	-0.80	-0.47	-1.08	-0.25	-0.83
Rotterdam	-1.69	-1.40	-0.29	-1.47	-1.00	-0.47
Dordrecht	-0.88	-0.67	-0.21	-0.69	-0.29	-0.40
Goes	-1.12	-0.96	-0.16	-1.08	-1.04	-0.04
Middelburg	-0.99	-0.64	-0.34	-0.98	-0.75	-0.24
Terneuzen	-1.30	-1.07	-0.24	-0.80	-0.36	-0.44
Breda	-0.78	-0.48	-0.31	-0.64	-0.52	-0.12
Tilburg	-1.41	-1.09	-0.32	-1.28	-0.72	-0.55
's-Hertogenbosch	-0.78	-0.76	-0.03	-0.67	-0.67	0.00
Eindhoven	-1.19	-1.02	-0.16	-1.05	-0.96	-0.09
Venlo	-1.04	-1.03	-0.01	-0.75	-0.17	-0.59
Roermond	-0.73	-0.90	0.17	-0.66	-0.83	0.16
Sittard	-0.70	-0.62	-0.08	-0.38	-0.38	0.00
Heerlen	-1.12	-0.99	-0.13	-0.74	-0.80	0.06
Maastricht	-2.27	-1.60	-0.67	-1.85	-0.99	-0.86

the WGR region). Figures 6 and 7 show these relationships for employment. The X axis in these figures presents the difference between morphological and functional polycentricity expressed in percentage point differences.

In other words, those regions that are positioned to the right, with a score greater than zero, are more morphologically polycentric than functionally polycentric. The opposite holds for most regions. Their

Table 2. Correlation matrix of the different dimensions of the spatial organisation of WGR regions

	(1)	(2)	(3)	(4)
<i>Employment</i>				
Morphological polycentricity (1)	1.00			
Functional polycentricity (2)	0.84	1.00		
Network density (3)	0.30	0.10	1.00	
Distribution of external centrality (4)	0.90	0.78	0.18	1.00
<i>Shopping</i>				
Morphological polycentricity (1)	1.00			
Functional polycentricity (2)	0.57	1.00		
Network density (3)	0.41	-0.01	1.00	
Distribution of external centrality (4)	0.55	0.47	0.14	1.00

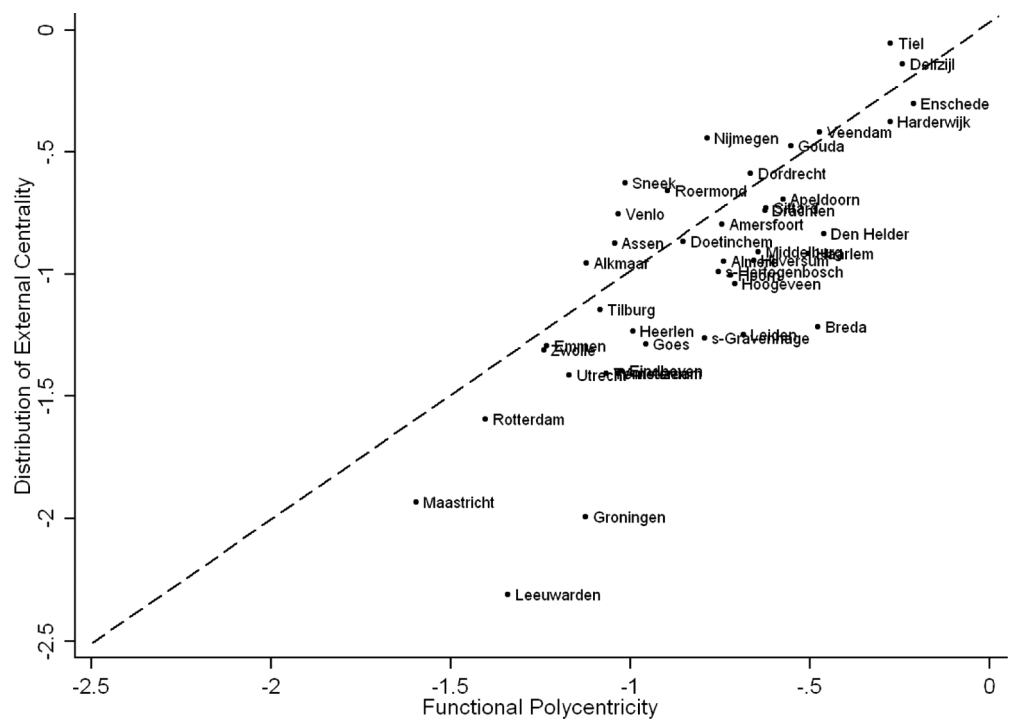


Figure 5. External centrality and functional polycentricity in WGR regions (employment).

negative score indicates that these are more functionally polycentric than morphologically polycentric. The further to the left the score is, the larger this discrepancy is.

Figure 6 shows that regions that are more functionally polycentric than morphologically polycentric tend to have a principal city that

draws more heavily on the local population, in our case the local labour market. Obviously it is easier to match labour demand and supply locally when the local labour market is larger.

Figure 7 shows that when the principal city in a region has stronger external linkages with places outside the region, it is likely to be more

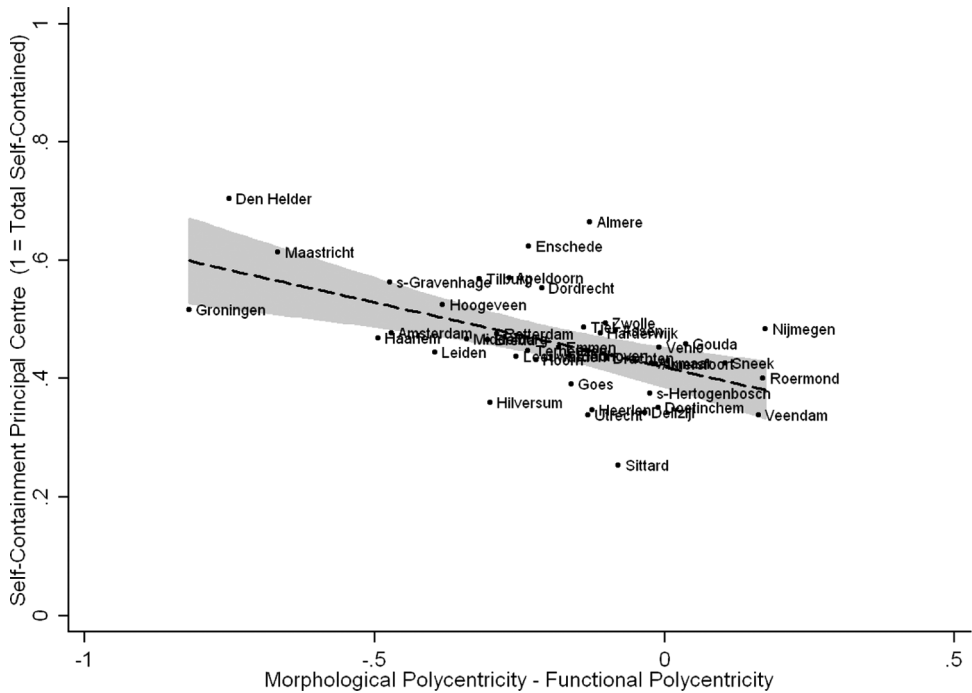


Figure 6. Local orientation principal centre and the difference between morphological and functional polycentricity: employment.

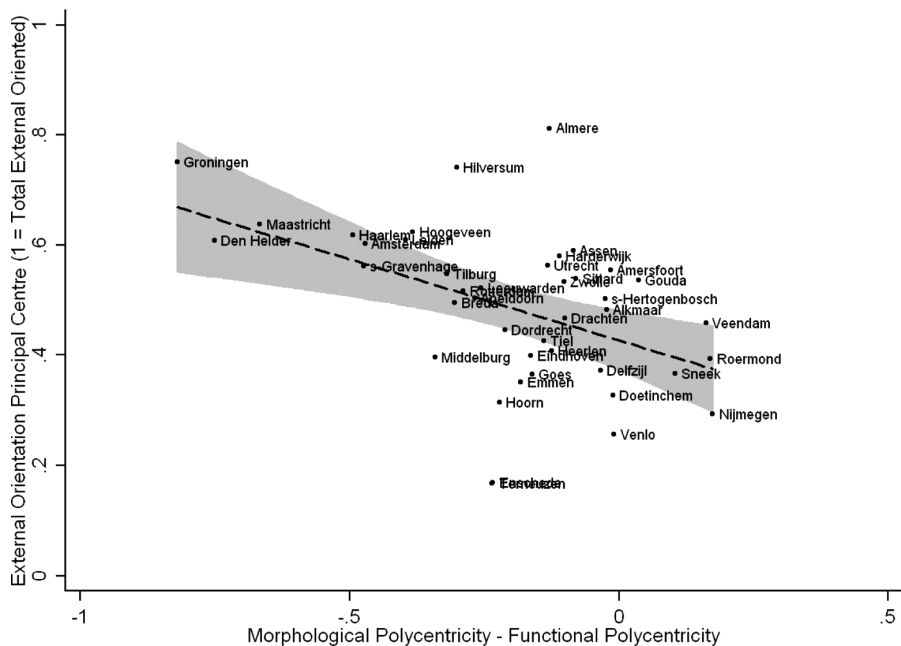


Figure 7. External orientation principal centre and the difference between morphological and functional polycentricity: employment.

functionally polycentric than morphologically polycentric. In our case, principal cities that draw more commuters from outside the WGR region tend to be located in regions that are more functionally polycentric than morphologically polycentric.

Concerning differences between the regions, regions that are substantially more functionally polycentric than morphologically polycentric have principal centres that are large in absolute terms (for example, Amsterdam, 's-Gravenhage (The Hague), Groningen, Maastricht). Conversely, regions that are relatively more morphologically polycentric than functionally polycentric (for example, Roermond and Veendam) have a relatively small primary centre that is subordinate in the supraregional urban system (Figure 8).

5.3 Regional Variations

Finally, we want to show how our four measures (nodality, internal centrality, external

centrality and local importance) relate to each other at the scale of individual WGR regions. In this, we focus again on employment. We present the WGR regions Groningen, Utrecht and Veendam in Figure 9. Groningen is an example of a region that is much more morphologically polycentric than functionally polycentric. On the other hand, Veendam is one of the few regions that is more functionally polycentric than morphologically polycentric, whereas for Utrecht there is hardly any difference between the degree of morphological and functional polycentricity. For Groningen, it is obvious that nodality scores (related to morphological polycentricity) are distributed in a less balanced way than the centrality scores of the four largest places in the region (related to functional polycentricity). External centrality, however, has a slightly more skewed distribution than morphological polycentricity. In other words, the largest city, Groningen, maintains relatively more

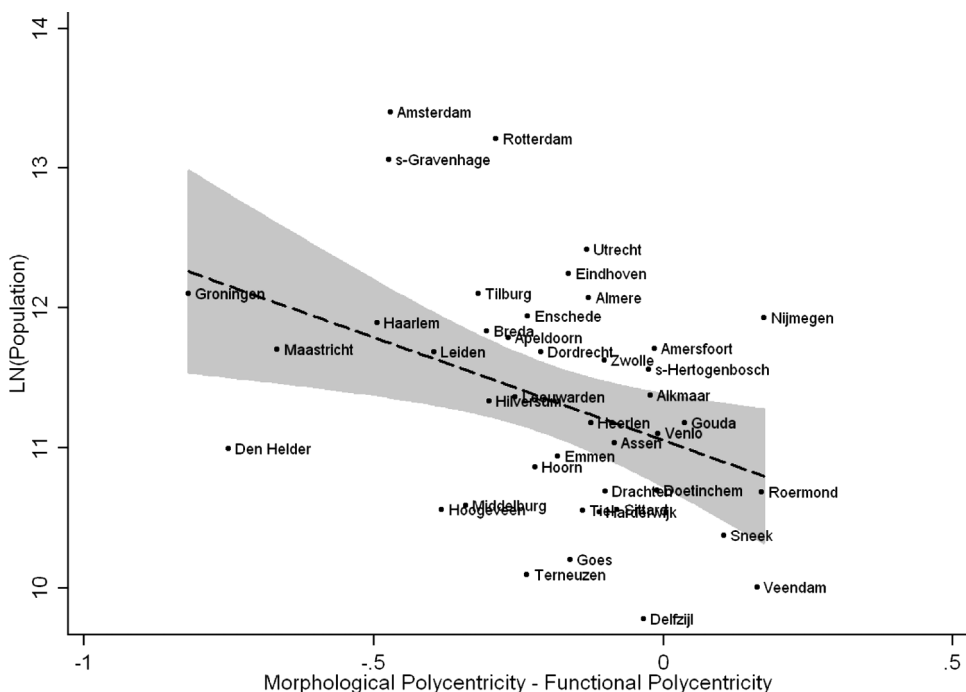


Figure 8. Principal centre size and the difference between morphological and functional polycentricity: employment.

relations (flows) with places from outside the region than we would expect given the size (nodality) distribution, whereas the lower-ranked cities in this region are more oriented towards other places within the region. It

is also clearly visible that local importance is distributed more unevenly than nodality and centrality. A much larger percentage of jobs (54 per cent) in the city of Groningen is filled by workers who also live in the city

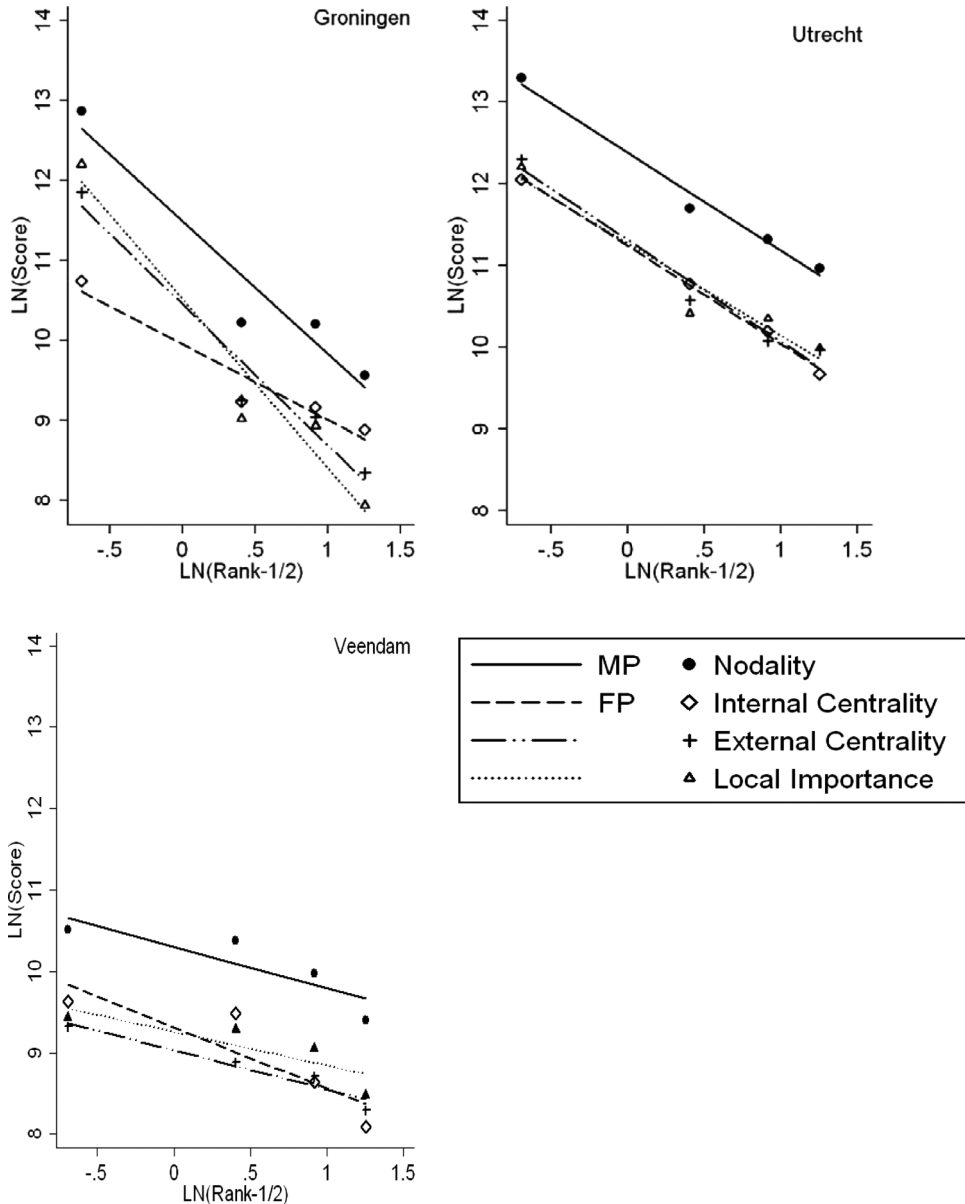


Figure 9. Morphological polycentricity (MP), functional polycentricity (FP), the distribution of external centrality and the distribution of local importance in 3 prototypical WGR regions: employment.

than is the case with the other, lower-ranked cities in the Groningen region (24 per cent). These are less able to draw workers from their own local labour market. Utrecht is hardly more functionally polycentric than morphologically polycentric. This tends to come coupled with reasonably similar distributions of local importance, internal centrality and external centrality. Comparing the Utrecht and Groningen WGR regions, it can be seen from the less steep slopes for Utrecht, that Groningen is in all respects more monocentric than Utrecht. This does not hold for the region in which Veendam is the largest centre. This region is one of the most morphologically polycentric regions in the Netherlands. Yet, this does not automatically imply that internal commuting flows are evenly distributed to the same extent.

6. Discussion and Conclusions

The lack of conceptual clarity surrounding the fuzzy concept of polycentricity has long impeded the much needed and often called for progress in our knowledge of the actual merits of polycentricity and the need for polycentric development policies. However, many of the contributions to the debate on polycentricity over recent years have highlighted the variety of interpretations and approaches towards the concept of polycentricity (for example, Lambregts, 2009, provides an excellent overview) and, therefore, this paper aims to shed light on what can be considered as the next step in this debate: the measurement of polycentricity. Not surprising given the variety in approaches to polycentricity, there is no consensus on what to measure. We identified, in the literature, two dominant but analytically distinct approaches. The first one, often referred to as morphological polycentricity, basically addresses the size of the urban centres across the territory and equates more balanced distributions with polycentricity. The second

approach takes relations between the centres into account and is referred to as functional polycentricity. A balanced, multidirectional set of relations between urban centres is considered more polycentric. Rather than taking a normative stance towards one approach or the other, we show that both approaches share the same basic principle in that both are concerned with the balance in importance of urban centres in a given area. This enables a similar method of measurement to be used and hence enables a comparison of morphological and functional polycentricity. Informed by the rich heritage of central place and urban systems research, this paper presents a model that links both approaches and discusses the way both can be measured and compared. We provide this comparison for 42 functionally coherent regions in the Netherlands. To enhance robustness, we did these analyses using employment (commuting) and shopping data. The following conclusions can be drawn

- There is no dominant type of spatial organisation in the regions. Some are monocentric, some polycentric and most are somewhere in between.
- Despite a considerable correlation between the degree of morphological and functional polycentricity of the regions, almost all the regions are relatively more functionally polycentric than morphologically polycentric.
- The greater this dominance of functional polycentricity over morphological polycentricity, the greater the degree to which the principal city is self-sufficient, building on its own local labour and consumer market; the more flows the principal city attracts from places from outside its own region; and the larger the size of the principal city.

Hence, large differences between the degree of morphological and functional polycentricity of regions come coupled with a relatively large

principal centre that has both a stronger local *and* external orientation. This can be explained by the fact that this difference also increases the larger the principal city is. Centre size is positively associated with sectoral diversity and a diverse occupational mix (Jacobs, 1969; Duranton and Puga, 2000). Size also brings with it a larger local labour force, enabling a better match between labour supply and demand. Moreover, higher-order functions (including specialised retail establishments) are still more often found in larger cities (Ross, 1992; Glaeser *et al.*, 2001; Markusen and Schrock, 2006). This makes principal centres more self-sufficient than the lower-order centres. In addition, the over-representation of higher-order functions in the principal centres may also attract a disproportionate number of people from outside the region. In this, it is well known that higher-ranked employees (in terms of education and income) are willing to commute longer distances to work (Schwanen and Dijst, 2002) and consumers are willing to travel longer distances to purchase specialised goods and services (Dijst and Vidakovic, 2000). As these explanations also hold outside the Netherlands, it is likely that we shall find similar results for regions in other countries or at different scales, such as countries or cross-border macro regions. Yet, one has to be aware that the Netherlands is a comparatively densely populated country and most of its cities are small or medium-sized, which might imply that general levels of polycentricity are relatively high, while it could be assumed that the external centrality of a region's principal city remains relatively low. Therefore, explorations for other countries will reveal whether these results can be generalised.

This paper has taken commuting and shopping as the primary features upon which to build our analysis. We may reflect on the consequences of taking other flow data. Although the effect of the distribution of external centrality on the difference between morphological and functional polycentricity

is rather limited, it can be expected that when assessing the spatial structure of territories on the basis of interfirm trade or shareholder relations, the external centrality of centres would play a more important role as the geographical scope of these functional linkages is usually also larger.

It is our hope that this contribution opens up a research agenda on polycentricity that is no longer dominated by conceptual issues, but that focuses on whether the alleged benefits of polycentricity and polycentric development hold true or not. Such an evidence base is necessary to determine whether polycentric development as a policy concept is sustainable. In actual fact, such research on the relationship between polycentricity and regional performance is of pivotal importance, given that polycentric development is a key policy concept in discussions of territorial cohesion (a potential third pillar of cohesion policy next to economic and social cohesion) and that considerable amounts of public investment could accordingly be spent in sub-optimal ways. This paper suggests that, in carrying out this research, it is essential to distinguish between morphological and functional polycentricity, and that any associated benefits of these may be related to other characteristics of the urban system, such as the degree of network density or a region's capacity to draw in flows from further away.

Notes

1. A good overview of these models can be found in Berry and Parr (1988).
2. This does not mean that the study of spatial structure has been limited to central-place-based studies. Similarly, interrelated conceptualisations of hierarchical spatial structures can be found in other fields of research. Most notably, studies on metropolitan dominance (McKenzie, 1933; Duncan *et al.*, 1960), focused mainly on the structure of corporate networks and administrative hierarchies (see Ross, 1992,

for an overview), while studies drawing on graph theory and spatial interaction models—rooted in quantitative planning studies and regional science—have explicitly focused on the structure of the physical transport and communication networks, commuting and migration networks as well as intraregional trade (Isard, 1960; Kansky, 1963; Griffith, 1976).

3. See also Ross (1992), for a discussion of this issue in the field of urban sociology.
4. Yet, Preston (1971) already considered the centrality based on the consumption of people that neither live in the centre nor in the complementary region, which he labelled 'irregular consumption'. The reason why Christaller (1933) only focused on centrality based on the consumption of people living in the complementary region was the lack of mobility of people back in the 1930s.
5. This definition comes close to what Green (2007) labels 'ordinary functional polycentricity'.
6. See Limtanakool *et al.* (2007), de Goei *et al.* (2010) and Burger *et al.* (2011) for similar approaches.
7. It is in fact the only official recent delimitation of functionally coherent regions in the Netherlands and one of its advantages is that it is not by definition confined to traditional administrative borders.
8. The parameter values have been estimated using the rank–size regression approach of Gabaix and Ibragimov (2011), which corrects for small sample bias.
9. In this, we calculated the yearly average scores. In addition, scores were weighted so that they are representative for the whole Dutch population.
10. In the remainder, we focus predominantly on employment and commuting flows. Results for shopping are available on request.

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References

- Adolphson, M. (2009) Estimating a polycentric urban structure: case study: urban changes in the Stockholm region, *Journal of Urban Planning and Development*, 135, pp. 19–30.
- Albrechts, L. (2001) How to proceed from image and discourse to action: as applied to the Flemish Diamond, *Urban Studies*, 38, pp. 733–745.
- Barton, B. (1978) The creation of centrality, *Annals of the Association of American Geographers*, 68, pp. 34–44.
- Batten, D. F. (1995) Network cities: creative urban agglomerations for the 21st century, *Urban Studies*, 32, pp. 313–327.
- Beckmann, M. J. (1958) City hierarchies and the distribution of city sizes, *Economic Development and Cultural Change*, 6, pp. 243–258.
- Beckmann, M. J. and McPherson, J. (1970) City size distribution in a central place hierarchy: an alternative approach, *Journal of Regional Science*, 10, pp. 25–33.
- Berry, B. J. L. (1964) Cities as systems within systems of cities, *Papers of the Regional Science Association*, 13, pp. 146–163.
- Berry, B. J. L. and Garrison, W. L. (1958) The functional bases of the central place hierarchy, *Economic Geography*, 34, pp. 145–154.
- Berry, B. J. L. and Parr, J. B. with Epstein, B. J. *et al.* (1988) *Market Centres as Retail Locations*. Englewood Cliffs, NJ: Prentice Hall.
- Berry, B. J. L. and Pred, A. (1965) *Central place studies: a bibliography of theory and applications*. Regional Science Research Institute, Philadelphia, PA.
- Bertalanffy, L. von (1950) An outline of general systems theory, *The British Journal for the Philosophy of Science*, I(2), pp. 134–165.
- Bourne, L. S. and Simmons, J. W. (Eds) (1978) *Systems of Cities: Readings on Structure, Growth, and Policy*. New York: Oxford University Press.
- Burger, M., Goei, B. de, Laan, L. van der and Huisman, F. J. M. (2011) Heterogeneous development of metropolitan spatial structure: evidence from commuting patterns in English and Welsh city-regions, 1981–2001, *Cities*, 28, pp. 160–170.
- Champion, A. G. (2001) A changing demographic regime and evolving polycentric urban regions: consequences for the size,

- composition and distribution of city populations, *Urban Studies*, 38, pp. 657–677.
- Cheshire, P. (1999) Trends in sizes and structures of urban areas, in: P. Cheshire and E. S. Mills (Eds) *Handbook of Regional and Urban Economics*, Vol. 3, pp. 1339–1372. Amsterdam: Elsevier Science.
- Christaller, W. (1933) *Die Zentralen Orte in Süddeutschland*. Jena: Gustav Fischer.
- Coffey, W. J., Bourne, L. S., Randall, J. E. et al. (1998) Urban systems research: past, present and future: a panel discussion, *Canadian Journal of Regional Science*, 21, pp. 327–364.
- Cowell, M. (2010) Polycentric regions: comparing complementarity and institutional governance in the San Francisco Bay area, the Randstad and Emilia-Romagna, *Urban Studies*, 47, pp. 945–965.
- Davies, W. K. D. (1967) Centrality and the central place hierarchy, *Urban Studies*, 4, pp. 61–79.
- Davoudi, S. (2003) Polycentricity in European spatial planning: from an analytical tool to a normative agenda?, *European Planning Studies*, 11, pp. 979–999.
- Davoudi, S. (2007) Polycentricity: panacea or pipedream?, in: N. Cattán (Ed.) *Cities and Networks in Europe*, pp. 65–74. Esher: John Libbey Eurotext.
- Dijkstra, M. J. and Vidakovic, V. (2000) Travel time ratio: the key factor of spatial reach, *Transportation*, 27, pp. 179–199.
- Duncan, O. D., Scott, W. R., Lieberman, S. and Duncan, B. (1960) *Metropolis and Region*. New York: Wiley.
- Duranton, G. and Puga, D. (2000) Diversity and specialisation in cities: why, where and when does it matter?, *Urban Studies*, 27, pp. 533–555.
- ESPON 1.1.1 (2004) *Potentials for polycentric development in Europe*. Nordregio/ESPON Monitoring Committee, Stockholm/Luxembourg.
- Gabaix, X. and Ibragimov, R. (2011) Rank-1/2: a simple way to improve the OLS estimation of tail exponents, *Journal of Business Economics and Statistics*, 29, pp. 24–39.
- Glaeser, E. L., Kolko, J. and Saiz, A. (2001) Consumer city, *Journal of Economic Geography*, 1, pp. 27–50.
- Goei, B. de, Burger, M., Oort, F. van and Kitson, M. (2010) Functional polycentrism and urban network development in the greater south east UK: evidence from commuting patterns, *Regional Studies*, 44, pp. 1149–1170.
- Green, N. (2007) Functional polycentricity: a formal definition in terms of social network analysis, *Urban Studies*, 44, pp. 2077–2103.
- Green, N. (2008) City states and the spatial in-between, *Town and Country Planning*, May, pp. 224–231.
- Griffith, D. A. (1976) Spatial structure and spatial interaction: a review, *Environment and Planning A*, 8, pp. 731–740.
- Haggett, P. (1965) *Locational Analysis in Human Geography*. London: Edward Arnold.
- Haggett, P. and Chorley, R. J. (1967) *Network Analysis in Geography*. London: Edward Arnold.
- Hague, C. and Kirk, K. (2003) *Polycentricity scoping study*. Office of the Deputy Prime Minister, London.
- Hall, P. and Pain, K. (Eds) (2006) *The Polycentric Metropolis: Learning from Mega-city Regions in Europe*. London: Earthscan.
- Hoyler, M., Kloosterman, R. C. and Sokol, M. (2008) Polycentric puzzles: emerging mega-city regions seen through the lens of advanced producer services, *Regional Studies*, 42, pp. 1055–1064.
- Isard, W. (1960) *Methods of Regional Analysis*. Cambridge, MA: MIT Press.
- Jacobs, J. (1969) *The Economy of Cities*. London: Jonathan Cape.
- Kansky, K. J. (1963) *Structure of transportation networks: relationships between network geometry and regional characteristics*. Research Paper No. 84, Department of Geography, University of Chicago, IL.
- Kloosterman, R. C. and Lambregts, B. (2001) Clustering of economic activities in polycentric urban regions: the case of the Randstad, *Urban Studies*, 38, pp. 717–732.
- Kloosterman, R. C. and Musterd, S. (2001) The polycentric urban region: towards a research agenda, *Urban Studies*, 38, pp. 623–633.
- Krätke, S. (2001) Strengthening the polycentric urban system in Europe: conclusions from the ESPD, *European Planning Studies*, 9, pp. 105–116.
- Laan, L. van der (1998) Changing urban systems: an empirical analysis at two spatial levels, *Regional Studies*, 32, pp. 235–247.

- Lambregts, B. (2009) *The polycentric metropolis unpacked: concepts, trends, and policy in the Randstad Holland*. Amsterdam Institute for Metropolitan and International Development Studies.
- Limtanakool, N., Schwanen, T. and Dijst, M. (2007) A theoretical framework and methodology for characterising national urban systems on the basis of flows of people: evidence for France and Germany, *Urban Studies*, 44, pp. 2123–2145.
- Lösch, A. (1944) *Die Räumliche Ordnung der Wirtschaft*. Jena: Gustav Fischer.
- Lukermann, F. (1966) Empirical expressions of nodality and hierarchy in a circulation manifold, *East Lakes Geographer*, 2, pp. 17–44.
- Markusen, A. (2003) Fuzzy concepts, scanty evidence, policy distance: the case for rigour and policy relevance in critical regional studies, *Regional Studies*, 37(6/7), pp. 701–717.
- Markusen, A. and Schrock, G. (2006) The distinctive city: divergent patterns in growth, hierarchy and specialisation, *Urban Studies*, 43, pp. 1301–1323.
- Marshall, J. U. (1989) *The Structure of Urban Systems*. Toronto: University of Toronto Press.
- McKenzie, S. (1933) *The Metropolitan Community*. New York: Russell & Russell.
- McPherson, J. C. (1981) The implications of central place theory for urban structure in a declining region: the North American experience, *Papers of the Regional Science Association*, 47, pp. 35–43.
- Meijers, E. (2005) Polycentric urban regions and the quest for synergy: is a network of cities more than the sum of its parts?, *Urban Studies*, 42, pp. 765–781.
- Meijers, E. (2007) From central place to network model: theory and evidence of a paradigm change, *Tijdschrift voor Economische en Sociale Geografie*, 98, pp. 245–259.
- Meijers, E. (2008a) Summing small cities does not make a large city: polycentric urban regions and the provision of cultural, leisure and sports amenities, *Urban Studies*, 45, pp. 2323–2342.
- Meijers, E. (2008b) Measuring polycentricity and its promises, *European Planning Studies*, 16, pp. 1313–1323.
- Meijers, E. and Burger, M. (2010) Spatial structure and productivity in US metropolitan areas, *Environment and Planning A*, 42, pp. 1383–1402.
- Meijers, E. and Sandberg, K. (2008) Reducing regional disparities by means of polycentric development: panacea or placebo?, *Scienze Regionali*, 7, pp. 71–96.
- Meijers, E., Waterhout, B. and Zonneveld, W. A. M. (2007) Closing the GAP: territorial cohesion through polycentric development, *European Journal of Spatial Development*, 24(October) (<http://www.nordregio.se/ejsd/refereed24.pdf>).
- Nystuen, J. D. and Dacey, M. F. (1961) A graph theory interpretation of nodal regions, *Papers of the Regional Science Association*, 7, pp. 29–42.
- Oort, F. van, Burger, M. and Raspe, O. (2010) On the economic foundation of the urban network paradigm: spatial integration, functional integration and economic complementarities within the Dutch Randstad, *Urban Studies*, 47, pp. 725–748.
- Parr, J. B. (1969) City hierarchies and the distribution of city size: a reconsideration of Beckmann's contribution, *Journal of Regional Science*, 15, pp. 1–8.
- Parr, J. B. (1987) Interaction in an urban system: aspects of trade and commuting, *Economic Geography*, 63, pp. 223–240.
- Parr, J. B. (2004) The polycentric urban region: a closer inspection, *Regional Studies*, 38, pp. 231–240.
- Parr, J. B. (2008) Cities and regions: problems and potentials, *Environment and Planning A*, 40, pp. 3009–3026.
- Phelps, N. A. and Ozawa, T. (2003) Contrasts in agglomeration: proto-industrial, industrial and post-industrial forms compared, *Progress in Human Geography*, 27, pp. 583–604.
- Preston, R. E. (1971) The structure of central place systems, *Economic Geography*, 47, pp. 136–155.
- Preston, R. E. (1975) A comparison of five measures of central place importance and of settlement size, *Tijdschrift voor Economische en Sociale Geografie*, 66, pp. 178–187.
- Ross, C. (1992) *The Urban System and Networks of Corporate Control*. Greenwich, CT: JAI Press.
- Schwanen, T. and Dijst, M. J. (2002) Travel-time ratios for visits to the workplace: the relationship between commuting time and work

- duration, *Transportation Research Part A*, 36, pp. 573–592.
- Scott, A. J., Agnew, J., Soja, E. and Storper, M. (2001) Global city-regions, in: A. J. Scott (Ed.) *Global City-regions: Trends, Theory, Policy*, pp. 11–30. Oxford: Oxford University Press.
- Short, J. R. (2004) Black holes and loose connections in a global urban network, *Professional Geographer*, 56, pp. 295–302.
- Simmons, J. W. (1978) The organization of the urban system, in: L. S. Bourne and J. W. Simmons (Eds) *Systems of Cities: Readings on Structure, Growth, and Policy*, pp. 61–69. New York: Oxford University Press.
- Thompson, D. (1974) Spatial interaction data, *Annals of the Association of American Geographers*, 64, pp. 560–575.
- Turok, I. and Bailey, N. (2004) The theory of polynuclear urban regions and its application to central Scotland, *European Planning Studies*, 12, pp. 371–389.
- Ullman, E. L. (1941) A theory of location for cities, *American Journal of Sociology*, 46, pp. 853–864.
- Vandermotten, C., Halbert, L., Roelandts, M. and Cornut, P. (2008) European planning and the polycentric consensus: wishful thinking?, *Regional Studies*, 42, pp. 1205–1217.
- Wall, R. S. (2009) The relative importance of Randstad cities within worldwide comparative corporate networks, *Tijdschrift voor Economische en Sociale Geografie*, 100, pp. 250–259.
- Wasserman, S. and Faust, K. (1994) *Social Network Analysis: Methods and Applications*. Cambridge: Cambridge University Press.
- Waterhout, B. (2002) Polycentric development: what is behind it, in: A. Faludi (Ed.) *European Spatial Planning*, pp. 83–103. Cambridge, MA: Lincoln Institute of Land Policy.
- Waterhout, B., Zonneveld, W. and Meijers, E. (2005) Polycentric development policies in Europe: overview and debate, *Built Environment*, 31, pp. 163–173.